TECHNICAL DATA SHEET

FR25 FIRE RATED UNDERLAYMENT

1. Product Name

TREADSTONE® FR25
FIRE RATED UNDERLAYMENT

2. Manufacturer

Formulated Materials, LLC
3010 NW 149th St., SUITE 100
Oklahoma City, OK 73134
Tech Services : (844) 405-3676

www.formulatedmaterials.com

www.formulated.info

3. Product Description

Treadstone® FR25 Fire Rated Underlayment sets a new standard in fire-resistive cementitious underlayments. Our innovative Treadstone® FR25 gypsum concrete adds the cementitious properties of, high strength, and abrasion resistance to gypsum underlayment's fire-resistive, non-shrinking, fast-cure properties. This gypsum concrete formula is mixed onsite with locally sourced sand and water to provide a dense, lightweight base that can host a wide range of flooring.

TREADSTONE® provides a full line of gypsum underlayments, self-levelers, and acoustic underlayments engineered to elevate the construction industry.

Key Features

- Lightweight: At 3/4" thick, it weighs 7.2 lbs. per sq. ft., reducing structural load.
- High Strength & Versatility: Mix designs available with compressive strengths ranging from 2,000 to 3,200 psi.
- Minimal Surface Prep: Simply prime with Treadstone® TP1 Primer/ Sealer and pour.
- Smooth, Dense Surface: Meets the most demanding floor covering requirements
- Fast Setting: Walkable in 4 hours, with regular trade traffic allowed within 24 hours.
- Adjustable Thickness: Pours from 3/8" to 3" in a single application; contact Formulated Materials for greater depth recommendations.
- Superior Sound Control: Enhances STC & IIC sound attenuation between floors.
- Eco-Friendly: Mixed on-site with locally sourced aggregates to reduce carbon footprint.
- Professional Installation: Installed by a technically trained network of licensed dealers.
- Code Compliant: UL listed in over 116 designs, ensuring fire resistance and safety compliance.

Suitable Substrates

- Properly prepared and/or primed wood, concrete, steel, gypsum, old tile adhesives, Treadstone® Technical Felt, or any Treadstone® Sound Attenuating Mats.

Product Composition

Gypsum-based formulation. Reinforced with proprietary additives for improved performance.

Product Limitations

- Do not exceed 3" lift per pour without consulting Formulated Materials.
- A moisture vapor barrier is required for installations over crawl spaces or high-MVER slabs.
- Not for exterior use or areas with long-term water exposure.
- Requires proper L/360 Deflection Criteria compliance for floor/ ceiling assemblies.
- FR25 is not a finished floor; a floor covering must be installed.
- Ensure all flanking paths are sealed to prevent sound transmission issues.
- Must be installed per Formulated Materials guidelines to ensure fire rating compliance.

These limitations are not to be construed as all inclusive.

Packaging

- 80 lb. bags / 49 bags per pallet / 2,960 lb. SuperSacks.

4. Technical Data

Applicable Standards

ASTM Test	Standard
C472	Standard Test Methods for Physical Testing of Gypsum, Gypsum Plasters and Gypsum Concrete
E84	Standard Test Method for Surface Burning Characteristics of Building Materials
E119	Standard Test Method for Fire Tests of Building Construction and Materials
F710	Standard Practice for Preparing Concrete Floors to Receive Resilient Flooing

UL Fire Rated Assemblies Compliance

Technical Chart

Property	Test Method	Results
Appearance		Powder
Color		Grey
Flow		High Flow
Water per 80# Bag		4.5 – 5.5 U.S. Gal.
Sand Per 80# Bag		1.4 to 2.1 cu. Ft. of Washed Mason or Concrete Sand meeting Formulated Materials Min. Gradation Criteria
Strength	ASTM C472	2000 – 3200 psi
Dry Density		110 – 120 lbs. per cu. ft.
Recommended Depths		3/8" – 3.0" Single Pour
Surface Burning Characteristics	ASTM E84	Flame Spread: 0, Fuel Contribution; 0, Smoke Density: 0
Fire Rated Assemblies		Refer to UL Fire Resistive Directory Listed in 116 designs

Environmental Considerations

FORMULATED MATERIALS is dedicated to environmental responsibility in product development and manufacturing processes. When used per applicable guidelines, this product may contribute to LEED® certification.

5. Instructions

General Surface Prep

Ensure the surface is clean, dry and structurally sound. Repair any cracks, spalling, or unstable areas. Remove dust, dirt, oil, grease, and contaminants. Use mechanical methods (grinding, sanding) if needed. Verify moisture levels meet product requirements. Address excessive moisture if necessary. Apply primer as specified and allow it to dry thoroughly. Maintain recommended temperature and humidity. Avoid extreme conditions. Perform a final check to clear debris and confirm the surface meets all preparation guidelines before application.

Installation of Product

Always read the Safety Data Sheet (SDS) for full precautionary instructions before use.

Tools Required:

- Mixing station with calibrated measuring equipment.
- SmartBatch® Pump (or equivalent mixing system).
- Skid-steer bucket (for sand measurement and loading).
- Clean water source (potable water only).
- Slump test kit (or 2" PVC pipe, plexiglass sheet, ruler, and mixing spatula).
- Screed or gauge rake (for leveling).
- Moisture meter (Delmhorst G-79 or BD 2000 recommended).
- Cube mold set (for strength testing per ASTM C-472).
- WD-40 (for lubricating cube molds).
- Sample containers (for water and sand testing).
- Paper towels and cleaning supplies.

Surface Prep

1. Clean the Substrate: Remove all dirt, dust, debris, oil, and foreign substances from the substrate.
2. Ensure Dry Conditions: The surface must be completely dry before application.
3. Apply Recommended Primer: Use the Treadstone® TP1 Primer/Sealer before pouring FR25.
4. Seal Substrate Penetrations: All openings, joints, and penetrations must be sealed or blocked to prevent leaks.
5. Comply with Dust Regulations: Avoid using oil-based floor sweeps as they can interfere with adhesion.
6. Check Moisture Vapor Emission: If moisture conditions exist, test per ASTM F-1869 to ensure MVER does not exceed 5 lbs per sq. ft. in 24 hours.

Application

Site Conditions

- The building must be enclosed and heated to a minimum of 50°F for at least 5-7 days before and after installation.
- Provide adequate ventilation to ensure proper curing. Consult Formulated Materials or refer to the Recommended Jobsite Conditions Document for guidance.
- Always follow the Floor Covering Manufacturer's recommendations.

Sand Selection

- Use well-graded, washed sand free of contaminants.
- Ensure sand meets the recommended sieve specifications.
- Perform an on-site Water Bottle Test to check for impurities.

Water Quality

- Use only clean, potable water from a municipal source.
- Avoid using water from ponds, wells, or untested sources.
- Adjust water quantity based on sand moisture content.

Mixing Instructions

- Measure Ingredients Accurately:
  - Follow mix design ratios (e.g., 1.8 cubic feet of sand per 80 lb bag of FR25).
  - Ensure water temperature remains between 45°F - 90°F.
- Batching Process:
  - Use a calibrated skid-steer bucket for consistent batching.
  - Follow Smart Batch pump procedures for accurate weight measurements.
- Ensure Proper Slump:
  - Optimal slump range for FR25: 8.5–9.5 inches.
  - Conduct multiple slump tests throughout the pour for consistency.

Pouring Instructions

- Pump mixed slurry onto the prepared substrate.
- Use a screed or gauge rake to distribute the material evenly.
- Monitor consistency to ensure proper thickness and uniformity.
- Allow material to settle before final smoothing.

Cube Sample Testing

- Prepare Sample Cubes:
  - Use a 1-quart container to collect material directly from the hose.
  - Keep the sample agitated until poured into the cube molds.
  - Lubricate cube molds with WD-40 before filling.
- Curing Cubes:
  - Leave undisturbed for 90-120 minutes.
  - Release cubes from the mold after full set.
- Submit Samples:
  - Package in an indestructo mailer box.
  - Ship to Formulated Materials with a completed Cube Set Form.
  - Testing performed per ASTM C-472

Curing & Protection

- Protect from Traffic: Prevent heavy foot or wheeled traffic until fully cured.
- Monitor Ambient Conditions:
  - Maintain minimum 50°F temperature until the underlayment is fully cured.
  - Provide continuous ventilation to remove excess moisture.

Storage & Handling

- Store in dry conditions.
- 12 month Shelf Life from date of manufacture when stored according to directions.

Health Precautions

- Follow all local safety regulations.
- Follow the manufacturer's Safety Data Sheets (SDS) and safety protocols for PPE.

Conformance to Building Codes

Installation must comply with all applicable local, state, and federal codes, regulations, and industry standards.

For additional technical support, contact Formulated Materials or visit FormulatedMaterials.com

6. Availability

Item	Description	Size
10100	FR25	80# Bag
10100-SS	FR25	2960# SuperSack

7. Product Warranty

For details and complete warranty information, visit www.formulatedmaterials.com

8. Product Maintenance

Properly installed products do not require ongoing maintenance.

9. Technical Services Information

For Technical Services, contact Formulated Materials at (844) 405-3676

10. Filing System

Additional product information is available from Formulated Materials upon request.

UL DESIGNS

G230	J920	L212	L515	L530	L550	L579	M508
G516	J924	L501	L516	L532	L551	L581	M511
G524	J927	L502	L517	L533	L552	L583	M512
G561	J931	L503	L518	L534	L556	L585	M513
G577	J957	L504	L519	L535	L557	L587	M515
G578	J958	L505	L520	L536	L558	L588	M517
G579	J991	L506	L522	L537	L560	L589	M519
G580	J994	L507	L523	L538	L562	L590	M530
G581	L006	L508	L523	L539	L563	L592	M535
G582	L201	L509	L524	L540	L565	L593	M536
G587	L202	L510	L525	L541	L570	M500	
G592	L206	L511	L526	L542	L573	M502	
H502	L209	L512	L527	L545	L574	M503	
J917	L210	L513	L528	L546	L576	M504	
J919	L211	L514	L529	L549	L577	M506	

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